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(54) **PRETAP EQUALIZABLE CONTINUOUS
TIME LINEAR EQUALIZER**

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H04L 27/01 (2006.01)

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CPC **H04L 25/03878** (2013.01); **H04L 1/0041**
(2013.01); **H04L 5/0005** (2013.01); **H04L**
27/01 (2013.01)

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5/0005; H04L 27/01
See application file for complete search history.

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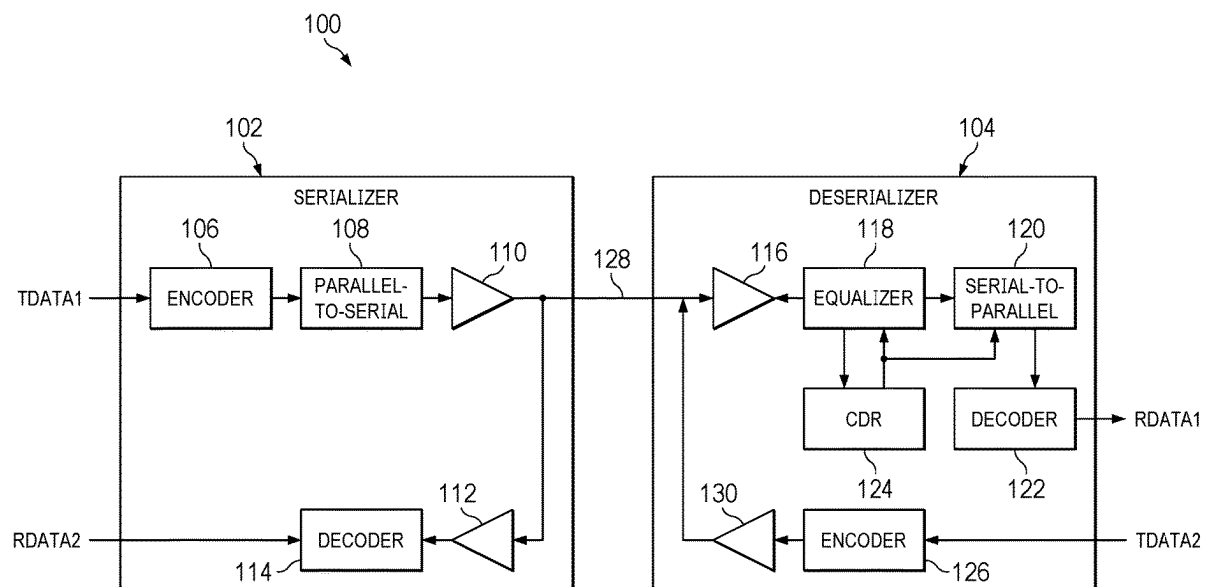
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(57) **ABSTRACT**

A circuit includes first, second, third, and fourth transistors, and a capacitor. The first transistor has a first terminal, a second terminal, and a control terminal. The second transistor has a first terminal, second terminal, and a control terminal. The capacitor has a first conductor coupled to the second terminal of the first transistor, and a second conductor coupled to the second terminal of the second transistor. The third transistor has a first terminal coupled to the first terminal of the second transistor, a second terminal, and a control terminal coupled to the control terminal of the first transistor. The fourth transistor has a first terminal coupled to the first terminal of the first transistor, a second terminal coupled to the second terminal of the third transistor, and a control terminal coupled to the control terminal of the second transistor.

20 Claims, 7 Drawing Sheets



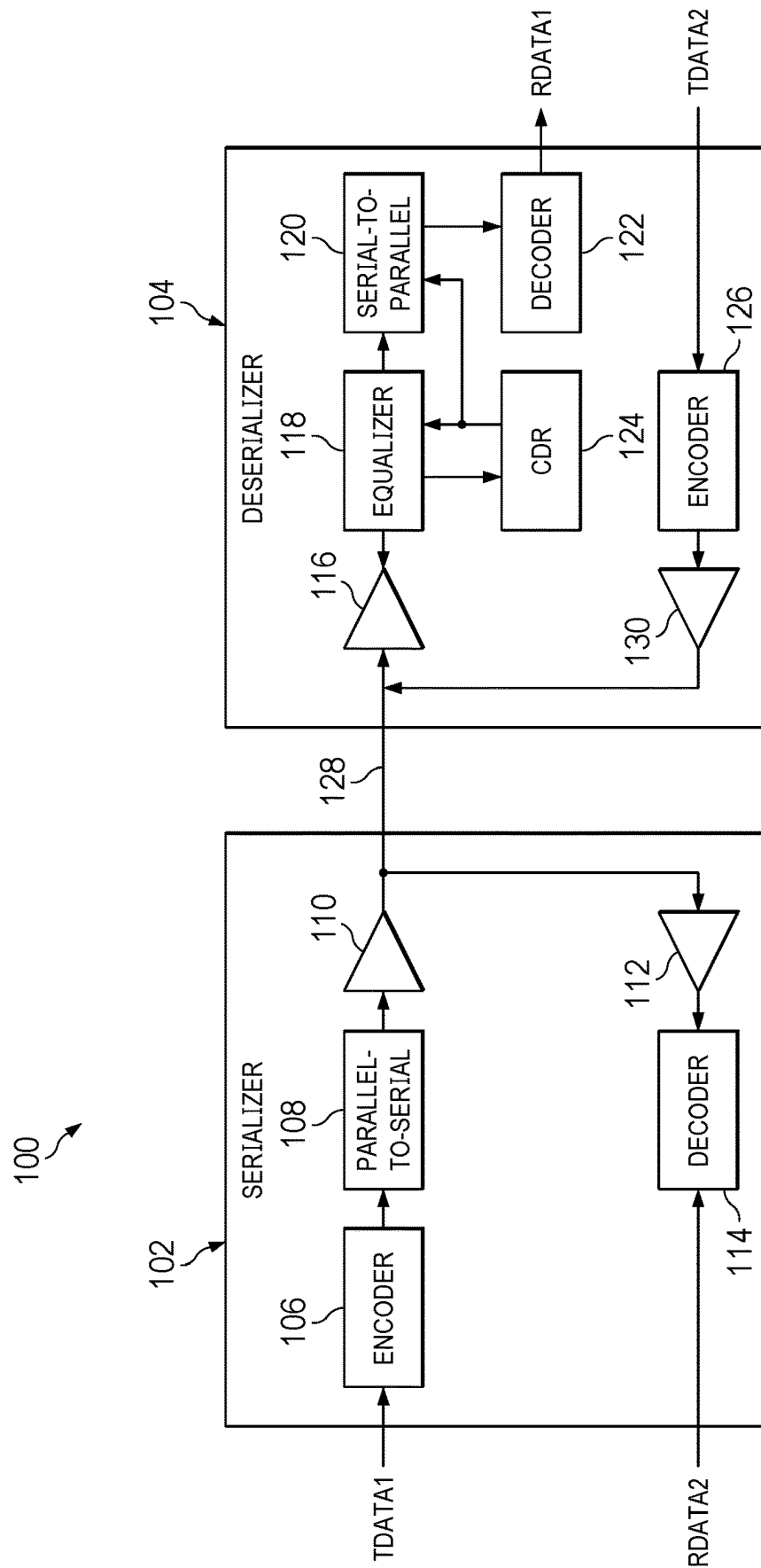


FIG. 1A

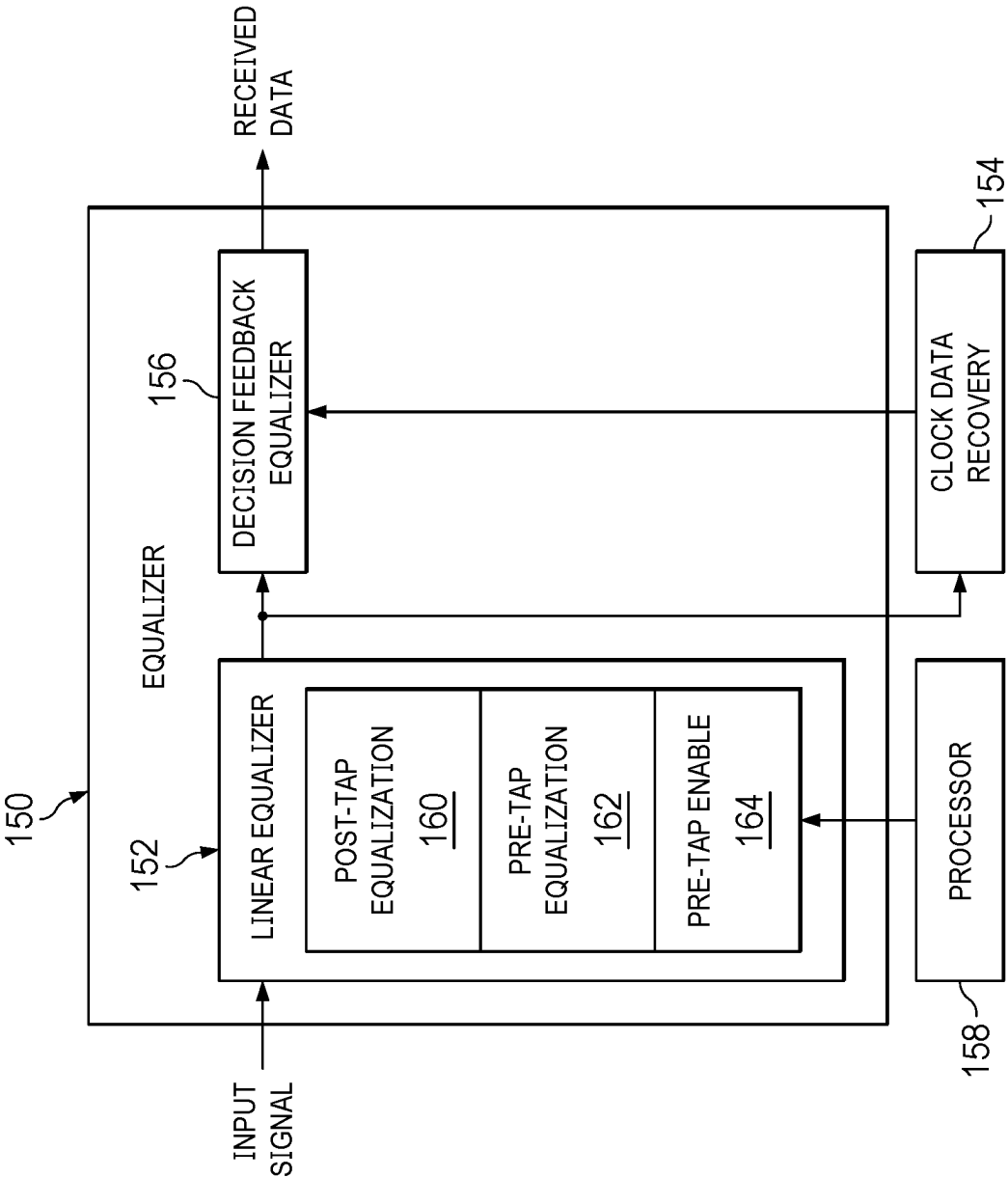


FIG. 1B

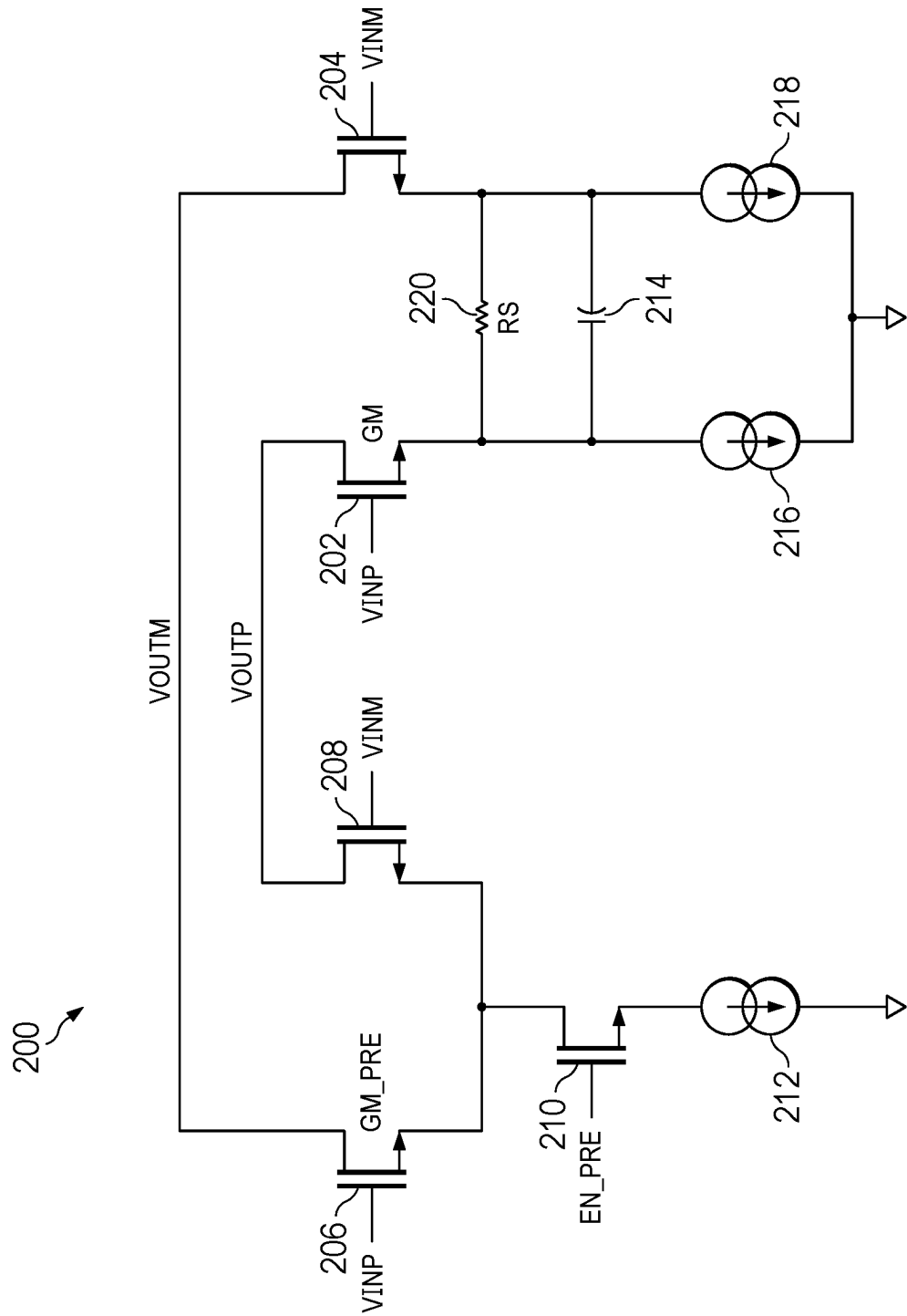


FIG. 2

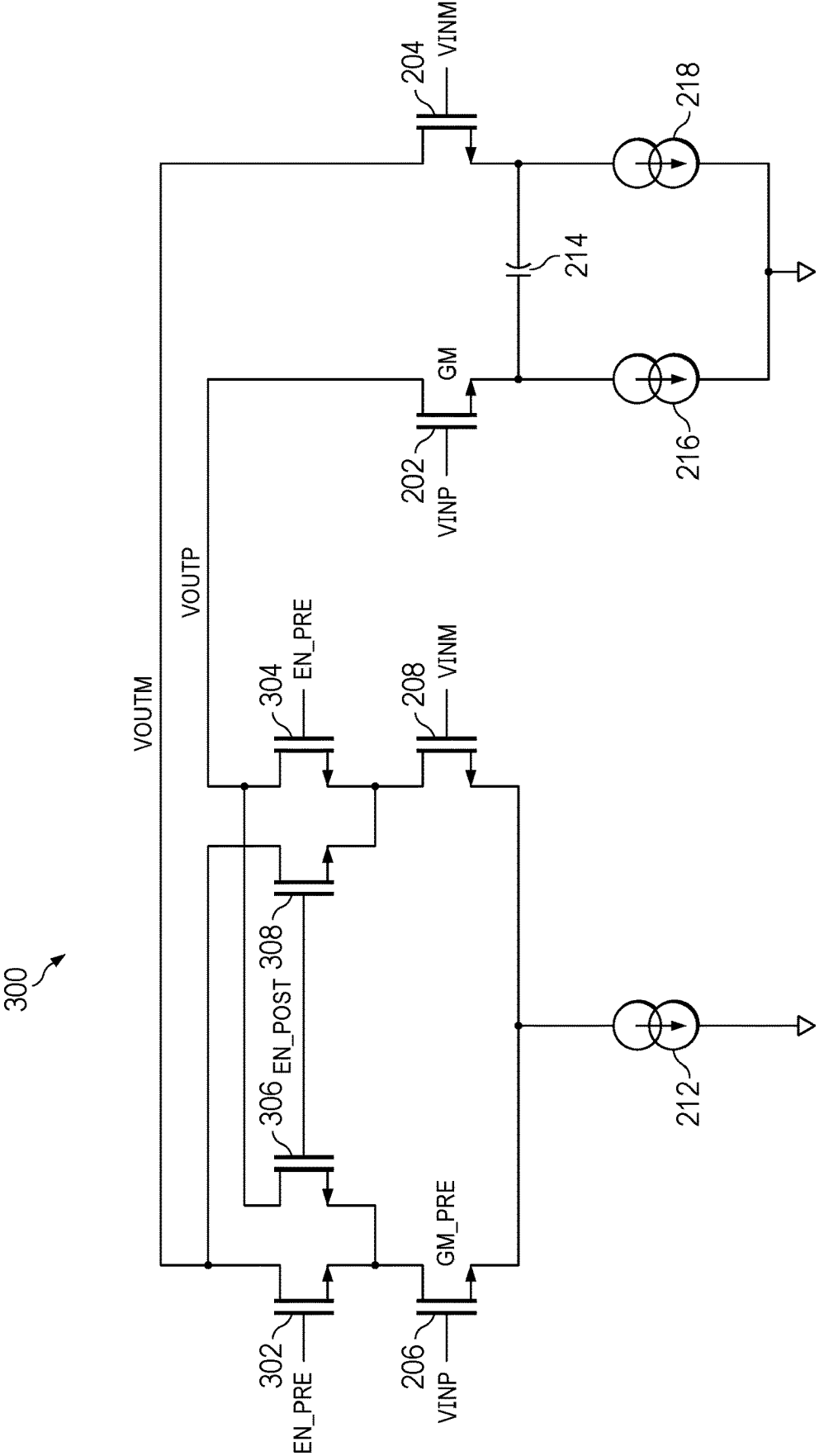
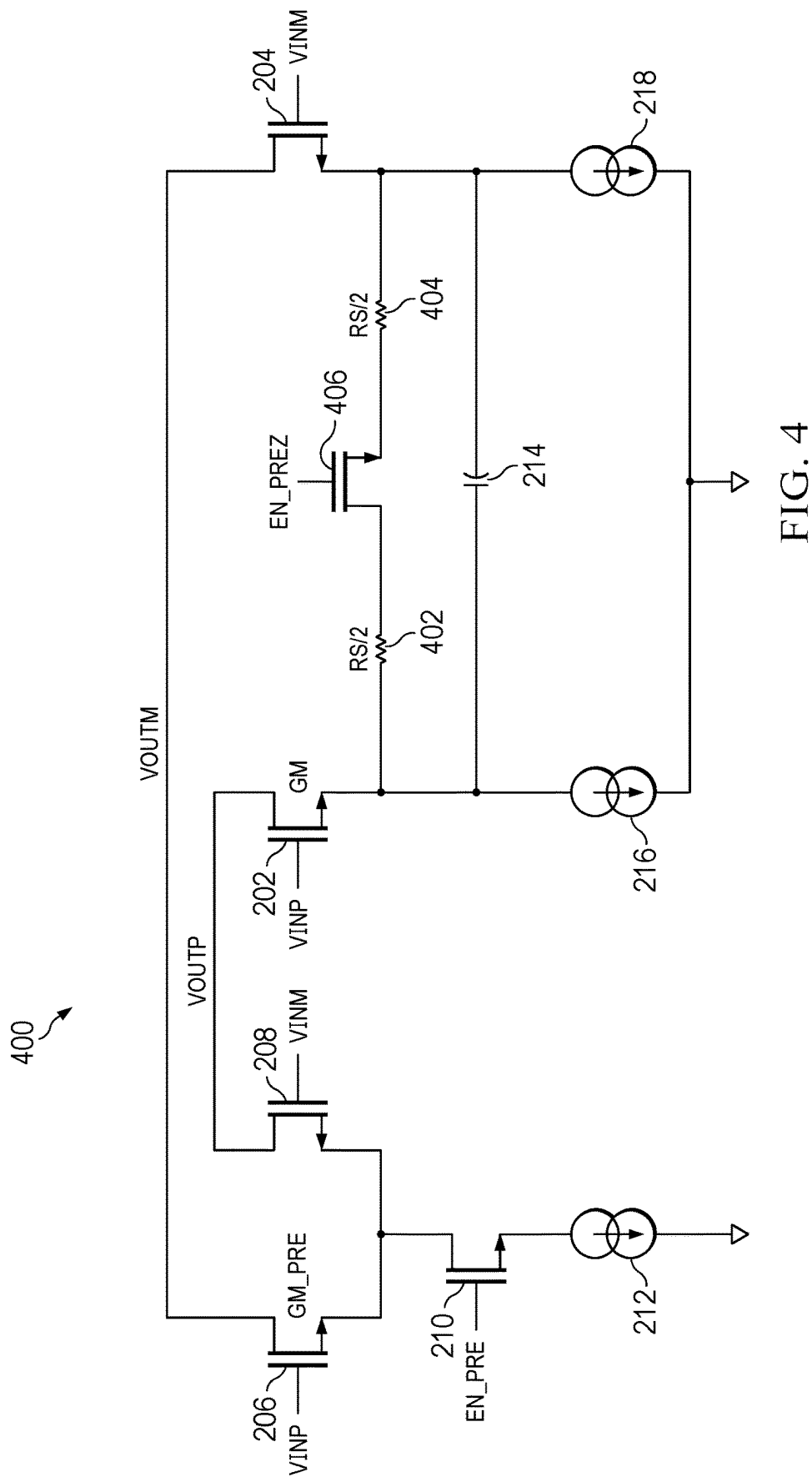


FIG. 3



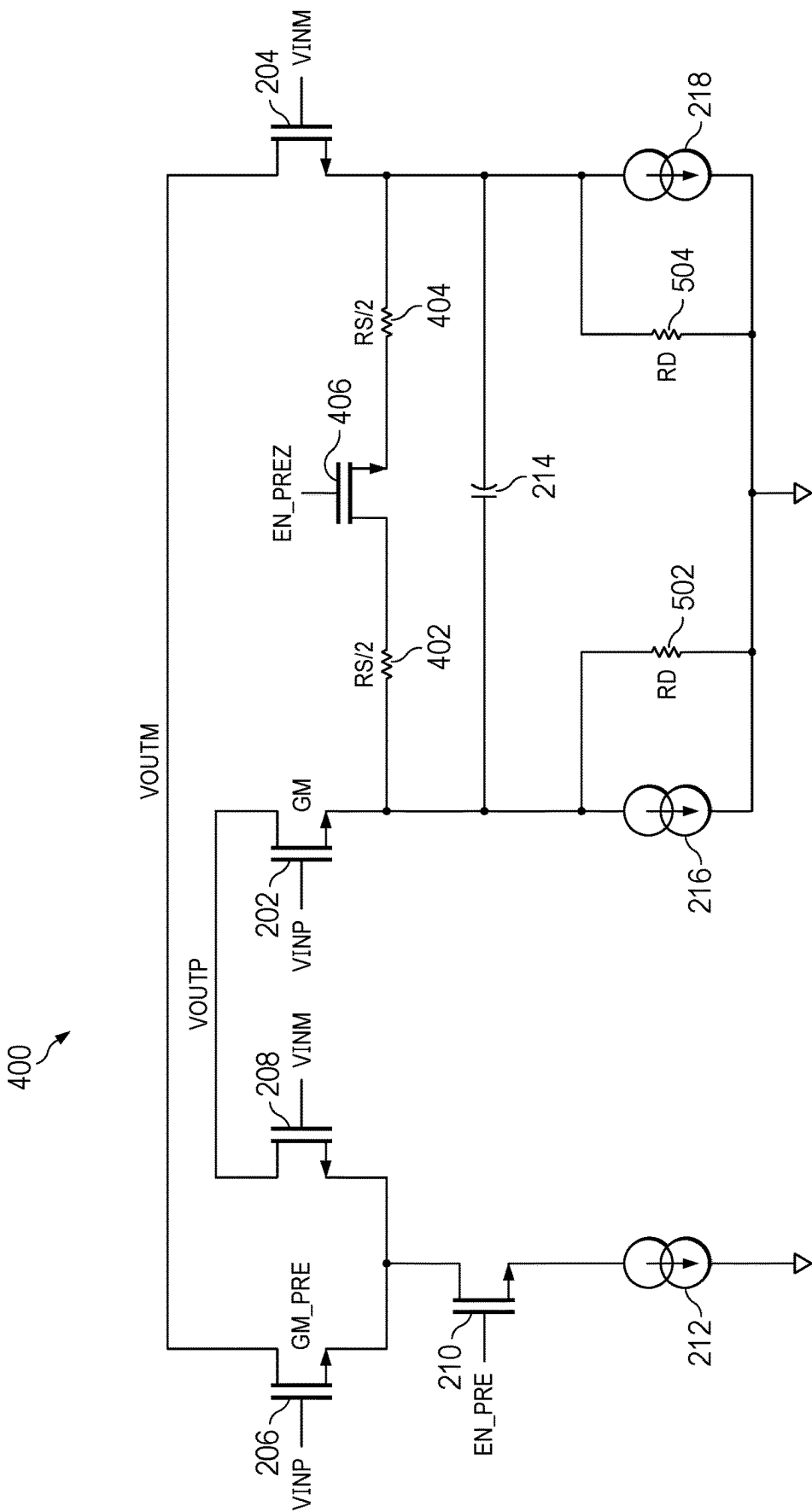


FIG. 5

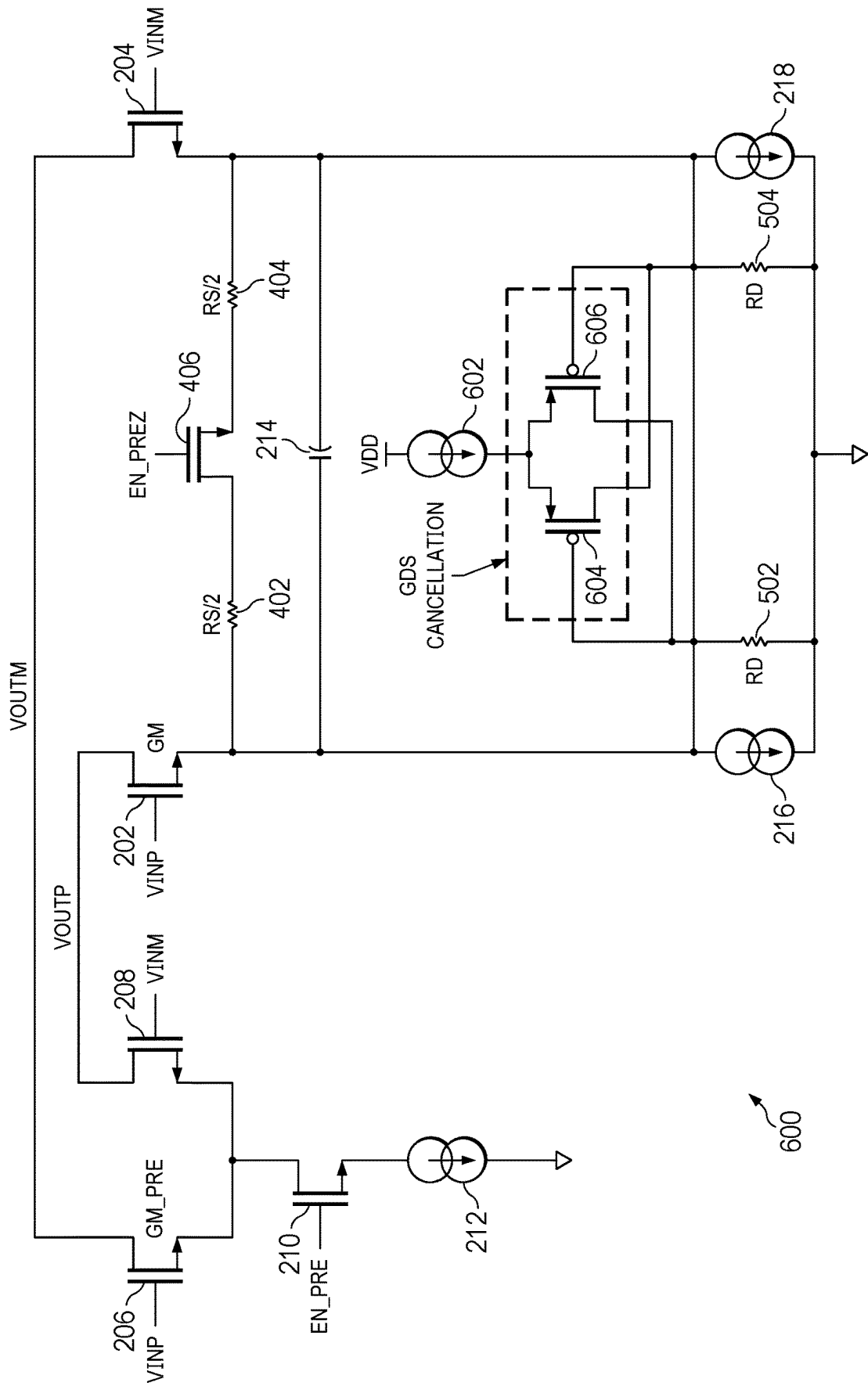


FIG. 6

1

PRETAP EQUALIZABLE CONTINUOUS TIME LINEAR EQUALIZER

BACKGROUND

Serial communication links experience high frequency distortion (phase and amplitude) between the transmitter and receiver over a lossy channel. This distortion is manifested at the receiver as inter-symbol interference, i.e., a smearing of the transmitted data bits/symbols. Channel equalization is used to counteract inter-symbol interference and other channel induced distortion. Channel equalization may be applied using transmission pre-emphasis that pre-distorts a transmit signal and/or as receiver equalization that applies post-compensation for the undesirable frequency effects of the channel.

SUMMARY

In one example, a circuit includes first, second, third, and fourth transistors, and a capacitor. The first transistor has a first terminal, a second terminal, and a control terminal. The second transistor has a first terminal, second terminal, and a control terminal. The capacitor has a first conductor coupled to the second terminal of the first transistor, and a second conductor coupled to the second terminal of the second transistor. The third transistor has a first terminal coupled to the first terminal of the second transistor, a second terminal, and a control terminal coupled to the control terminal of the first transistor. The fourth transistor has a first terminal coupled to the first terminal of the first transistor, a second terminal coupled to the second terminal of the third transistor, and a control terminal coupled to the control terminal of the second transistor.

In another example, a circuit includes first, second, third, fourth, and fifth transistors. The first transistor has a first terminal, a second terminal, and a control terminal. The second transistor is coupled to the first transistor as a first differential pair. The second transistor has a first terminal, a second terminal, and a control terminal. The third transistor has a first terminal, a second terminal, and a control terminal. The first terminal of the third transistor is coupled to the first terminal of the second transistor. The control terminal of the third transistor is coupled to the control terminal of the first transistor. The fourth transistor is coupled to the third transistor as a second differential pair. The fourth transistor has a first terminal, a second terminal, and a control terminal. The first terminal of the fourth transistor is coupled to the first terminal of the first transistor. The control terminal of the fourth transistor is coupled to the control terminal of the second transistor. The fifth transistor is coupled to the second differential pair. The fifth transistor is configured to switchably enable pre-tap equalization.

In a further example, a deserializer includes a clock data recovery circuit and a linear equalizer circuit. The linear equalizer circuit is coupled to the clock data recovery circuit. The linear equalizer circuit includes first, second, third, fourth, and fifth transistors, a capacitor, and first and second current sources. The first transistor has a first terminal, a second terminal, and a control terminal. The second transistor has a first terminal, second terminal, and a control terminal. The capacitor has a first conductor coupled to the second terminal of the first transistor, and a second conductor coupled to the second terminal of the second transistor. The first current source has a first terminal coupled to the first conductor, and a second terminal coupled to a reference voltage terminal. The second current source has a first

2

terminal coupled to the second conductor, and a second terminal coupled to the reference voltage terminal. The third transistor has a first terminal coupled to the first terminal of the second transistor, a second terminal, and a control terminal coupled to the control terminal of the first transistor. The fourth transistor has a first terminal coupled to the first terminal of the first transistor, a second terminal coupled to the second terminal of the third transistor, and a control terminal coupled to the control terminal of the second transistor. The fifth transistor is coupled to the third transistor. The fifth transistor is configured to switchably enable pre-tap equalization.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A a block diagram of an example serial communication system.

FIG. 1B is a block diagram of an example equalizer that includes linear equalization suitable for use in the deserializer of FIG. 1.

FIG. 2 is a schematic diagram of an example continuous time linear equalizer circuit with pre-tap equalization and optional post-tap equalization suitable for use in the equalizer of FIG. 1B.

FIG. 3 is a schematic diagram of a second example continuous time linear equalizer circuit with pre-tap equalization and optional post-tap equalization suitable for use in the equalizer of FIG. 1B.

FIG. 4 is a schematic diagram of a third example continuous time linear equalizer circuit with pre-tap equalization and optional post-tap equalization suitable for use in the equalizer of FIG. 1B.

FIG. 5 is a schematic diagram of the continuous time linear equalizer circuit of FIG. 4 showing parasitic resistances associated with the tail current sources.

FIG. 6 is a schematic diagram of a fourth example continuous time linear equalizer circuit with pre-tap equalization and optional post-tap equalization suitable for use in the equalizer of FIG. 1B.

DETAILED DESCRIPTION

FIG. 1A a block diagram of an example serial communication system **100**. The serial communication system **100** includes a serializer **102** and a deserializer **104** connected via a communication channel **128**. The communication channel **128** may be a wired connection, such as a coaxial cable or a twisted pair. The serializer **102** includes an encoder **106**, a parallel-to-serial converter **108**, and a transmitter **110** for forward channel communication. Forward channel communication may include transmission of video data, audio data, control data, etc. from the serializer **102** to the deserializer **104** via the communication channel **128**. Examples of the serializer **102** may also include a receiver **112** and a decoder **114** for back channel reception. The serializer **102** receives data to be transmitted (TDATA1), and the encoder **106** encodes the data for transmission. TDATA1 may be video received from video source or any other type of data provided by a data source. The encoder **106** may, for example, encode TDATA1 for DC balance, to add clock information, and/or introduce scrambling. The parallel-to-serial converter **108** converts the encoded data received from the encoder **106** to a serial bit-stream that is driven onto the communication channel **128** by the transmitter **110**. The transmitter **110** may provide differential or single-ended drive in various examples.

The deserializer **104** includes a receiver **116**, an equalizer **118**, a serial-to-parallel converter **120**, a decoder **122**, and a clock data recovery circuit **124** for forward channel reception. The deserializer **104** may also include an encoder **126** and a transmitter **130** for back channel transmission. The receiver **116** detects the signal transmitted by the serializer **102** propagated through the communication channel **128**. The equalizer **118** applies one or more equalization techniques to recover the data received from the communication channel **128**. The equalizer **118** provides equalized data to the serial-to-parallel converter **120** and the clock data recovery circuit **124**. The clock data recovery circuit **124** recovers a clock signal from the equalized data. The clock data recovery circuit **124** provides the recovered clock signal to the equalizer **118** and the serial-to-parallel converter **120**. For example, the equalizer **118** may apply the recovered clock signal to sample the equalized data. The serial-to-parallel converter **120** converts the equalized serial data to parallel symbols. The decoder **122** reverses the encoding applied in the serializer **102** to recreate the data received by the serializer **102** for transmission (to recreate TDATA1). Accordingly, the output of the decoder **122** (RDATA1) is a recreation of TDATA1. The deserializer **104** may provide RDATA1 to a video display, a video processing system, a processor, or any other circuitry configured to receive from the deserializer **104**.

Back channel communication may include transmission of control data or other data from the deserializer **104** to the serializer **102** via the communication channel **128** (e.g., simultaneous with forward channel transmission via the communication channel **128**). The encoder **126** encodes data (e.g., TDATA2) received for transmission. TDATA2 may be a serial data stream received from a processor or other device. The transmitter **130** drives the encoded data received from the encoder **126** onto the communication channel **128**.

In the serializer **102**, the receiver **112** receives the signal present on the communication channel **128** and separates the data transmitted by the transmitter **130** from the data transmitted by the transmitter **110**. The receiver **112** provides the received back channel data to the decoder **114**, and the decoder **114** reverses the encoding applied in the deserializer **104** to recreate the data received by the deserializer **104** for transmission (to recreate TDATA2). Accordingly, the output of the decoder **114** (RDATA2) is a recreation of TDATA2. RDATA2 may be provided to processor or other device.

FIG. 1B is a block diagram of an example equalizer **150**. The equalizer **150** may be an example of the equalizer **118**. The equalizer **150** includes a continuous time linear equalizer circuit **152**. Some examples of the equalizer **150** may also include a decision feedback equalizer **156**, a feed forward equalizer, or other equalizer circuit. A clock data recovery circuit **154** and a processor **158** may be coupled to the equalizer **150**. The continuous time linear equalizer circuit **152** is a linear equalizer that boosts both signal and noise. The decision feedback equalizer **156** is coupled to an output of the continuous time linear equalizer circuit **152**. The decision feedback equalizer **156** is a non-linear equalizer that is immune to noise, but may present feedback loop timing challenges. The clock data recovery circuit **154** extracts a clock signal from the output of the continuous time linear equalizer circuit **152** for use by the decision feedback equalizer **156** and/or other circuitry (e.g., the serial-to-parallel converter **120**).

The decision feedback equalizer **156** may provide post-tap equalization (correction of post-cursor inter-symbol interference). As data rate and modulation complexity increase, e.g., with implementation of pulse-amplitude

modulation 4-level (PAM4), pre-tap equalization (correction of pre-cursor inter-symbol interference) becomes desirable. In some systems, a feed-forward equalizer in discrete time may be used to provide pre-tap equalization. Discrete time feed-forward equalizers can be costly in terms of both power and circuit area. The equalizer **150** includes a post-tap equalization circuit **160**, a pre-tap equalization circuit **162**, and a pre-tap enable circuit **164**. The post-tap equalization circuit **160** provides post-tap equalization. The pre-tap equalization circuit **162** provides pre-tap equalization. The pre-tap enable circuit **164** controls the post-tap equalization circuit **160** and/or the pre-tap equalization circuit **162** to select pre or post-tap equalization based on a control signal received from the processor **158** or other control circuit. In the equalizer **150**, examples of the continuous time linear equalizer circuit **152** may provide pre-tap equalization with reduced power consumption and circuit area relative to other pre-tap equalization circuits.

The unit interval sampled pulse response of a continuous time linear equalizer output has a time domain response given by the finite impulse response:

$$H_{\text{post}}(z) = a - bZ^{-1} \quad (1)$$

where a is the coefficient of the main tap and b is the coefficient of the post tap.

The continuous time equivalent of $H(Z)$, ($H(s)$), can be found using inverse bilinear transformation:

$$Z = e^{sT} = \frac{1 + sT/2}{1 - sT/2} \quad (2)$$

where $T = 1/f_s$ (baud rate).

$$H_{\text{post}}(s) = (a - b) \left(\frac{1 + s \frac{(a+b)T}{2}}{1 + sT/2} \right) \quad (3)$$

Equation (3) shows that the left half plane zero generates a post-tap. The ratio of pole to zero

$$\left(\frac{(a+b)}{(a-b)} \right)$$

determines the boost in frequency and the values of the main tap coefficient a and the post-tap coefficient b .

To realize a pre-tap:

$$H_{\text{pre}}(z) = -b + aZ^{-1} \quad (4)$$

where the main tap has a coefficient a , and the pre-tap has a coefficient b .

Using bilinear transformation:

$$H_{\text{pre}}(s) = (a - b) \left(\frac{1 - s \frac{(a+b)T}{2}}{1 + sT/2} \right) \quad (5)$$

Comparing equations (3) and (5), the difference between H_{pre} and H_{post} is the presence of a right half plane zero in H_{pre} .

The post/pre-tap pulse response for a continuous time linear equalizer may be defined as:

$$H_{\text{ctle}}^{\text{pulse}}(t) = (U(t) - U(t-T)) \text{conv } H_{\text{ctle}}(t) \quad (6)$$

5

where $U(t)$ is unit step, T is pulse width corresponding to baud rate, and $H_{ctle}(t)$ is the impulse response of the continuous time linear equalizer.

In the Laplace domain:

$$H_{ctle_post_pulse}(s) = \left(\frac{1}{s} (1 - e^{-sT}) \right) X H_{post}(s) \quad (7)$$

$$\text{where } H_{post}(s) = \left(\frac{\left(1 + \frac{s}{Z}\right)}{\left(1 + \frac{s}{P}\right)} \right) \quad (8)$$

$$H_{ctle_post_pulse}(t) = \left[(e^{-Pt}) \left(\frac{P}{Z} - 1 \right) + 1 \right] u(t) - \left[(e^{-P(t-T)}) \left(\frac{P}{Z} - 1 \right) + 1 \right] u(t-T) \quad (9)$$

$$H_{ctle_pre_pulse}(s) = \left(\frac{1}{s} (1 - e^{-sT}) \right) X H_{pre}(s) \quad (10)$$

$$\text{where } H_{pre}(s) = \left(\frac{\left(1 + \frac{s}{Z}\right)}{\left(1 + \frac{s}{P}\right)} \right) \quad (11)$$

$$H_{ctle_pre_pulse}(t) = \left[1 - (e^{-Pt}) \left(\frac{P}{Z} + 1 \right) \right] u(t) - \left[1 - (e^{-P(t-T)}) \left(\frac{P}{Z} + 1 \right) \right] u(t-T) \quad (12)$$

A pre-tap continuous time linear equalizer may be derived from the post-tap continuous time linear equalizer of equation (8).

$$H_{pre}(s) = 2 - H_{post}(s) = 2 - \left(\frac{\left(1 + \frac{s}{Z}\right)}{\left(1 + \frac{s}{P}\right)} \right) \quad (13)$$

With zero $Z=P/K$, (boost of post tap continuous time linear equalizer= K)

$$H_{pre}(s) = \left(\frac{1 - \frac{s}{P}(k-2)}{\left(1 + \frac{s}{P}\right)} \right) \quad (14)$$

In equation (13), the right hand plane zero provides a pre-tap boost of $K-2$. $Z=P/(k+2)$ is needed to provide an equivalent post-tap continuous time linear equalizer boost.

FIG. 2 is a schematic diagram of an example continuous time linear equalizer circuit 200 with pre-tap equalization suitable for use in the equalizer 150. The continuous time linear equalizer circuit 200 is an example of the continuous time linear equalizer circuit 152. The continuous time linear equalizer circuit 200 includes transistors 202 and 204 connected as a first differential pair, and transistors 206 and 208 connected as a second differential pair. A first current terminal (e.g., drain) of the transistor 202 is coupled to a first output terminal (VOUTP) and to a first current terminal (e.g., drain) of the transistor 208. A first current terminal (e.g., drain) of the transistor 204 is coupled to a second output terminal (VOUTM) and to a first current terminal (e.g., drain) of the transistor 206. A control terminal (e.g., gate) of the transistor 202 is coupled to a first input terminal (VINP) and to a control terminal (e.g., gate of the transistor 206). A control terminal (e.g., gate) of the transistor 204 is coupled to a second input terminal (VINM) and to a control

6

terminal (e.g., gate of the transistor 208). A second current terminal (e.g., source) of the transistor 206 is coupled to a second control terminal (e.g., source) of the transistor 208.

The continuous time linear equalizer circuit 200 also includes a transistor 210, current sources 212, 216, and 218, a resistor 220, and a capacitor 214. A first current terminal (e.g., drain) of the transistor 210 is coupled to the second current terminals of the transistor 206 and the transistor 208. A control terminal (e.g., gate) of the transistor 210 is coupled to a pre-tap equalization enable terminal (EN_PRE). A second current terminal (e.g., source) of the transistor 210 is coupled to a first terminal of the current source 212. A second terminal of the current source 212 is coupled to a reference voltage terminal (e.g., ground). The transistor 210 may be turned on to switchably enable pre-tap equalization (activate tail current flow through the transistors 206 and 208) in the continuous time linear equalizer circuit 200, and turned off to disable pre-tap equalization. The transistors 202, 204, 206, 208, and 210 may be n-type field effect transistors.

A first terminal of the resistor 220 is coupled to a second current terminal (e.g., source) of the transistor 202. A second terminal of the resistor 220 is coupled to a second current terminal (e.g., source) of the transistor 204. The capacitor 214 is coupled to the transistor 202 and the transistor 204 in parallel with the resistor 220. A first conductor (e.g., top plate) of the capacitor 214 is coupled to the first terminal of the resistor 220, and a second conductor (e.g., bottom plate) of the capacitor 214 is coupled to the second terminal of the resistor 220.

The current source 216 is coupled between the second current terminal of the transistor 202 and the reference voltage terminal. The current source 218 is coupled between the second current terminal of the transistor 204 and the reference voltage terminal. A first terminal of the current source 216 is coupled to the second current terminal of the transistor 202, and a second terminal of the current source 216 is coupled to the reference voltage terminal. A first terminal of the current source 218 is coupled to the second current terminal of the transistor 204, and a second terminal of the current source 218 is coupled to the reference voltage terminal.

In the continuous time linear equalizer circuit 200, the transistor 210 is turned off (EN_PRE=0) to activate post-tap equalization and the transistor 210 is turned on (EN_PRE=1) to activate pre-tap equalization. If the transistor 210 is turned off,

$$H_{ctle_post}(s) = H_{ctle}(s) = \left(\frac{gm}{1 + gm * Rs / 2} \right) \left(\frac{1 + S(Rs * Cs)}{1 + \left(\frac{Rs}{1 + gm * \frac{Rs}{2}} \right)} \right) * Z_{out} \quad (14)$$

With

$$dcgain = \frac{gm}{1 + gm * Rs / 2}, \quad p = \frac{\left(1 + gm * \frac{Rs}{2}\right)}{Rs * Cs},$$

$$\text{and } z = \frac{(1)}{Rs * Cs} = \frac{P}{1 + gm * \frac{Rs}{2}},$$

7

the post boost is

$$K = 1 + gm * \frac{Rs}{2} \quad (15)$$

If the transistor **210** is turned on, the transistor **206** and the transistor **208** turn on with:

$$GM_PRE = 2 * \left(\frac{gm}{1 + gm * Rs / 2} \right) \quad (16)$$

$$H_{ctle_pre}(s) = \quad (17)$$

$$\left(\frac{gm}{1 + gm * Rs / 2} \right) \left(\frac{1 - \frac{S * Rs * Cs}{1 + gm * \frac{Rs}{2}} * \left(gm * \frac{Rs}{2} - 1 \right)}{1 + \frac{S * Rs * Cs}{1 + gm * \frac{Rs}{2}}} \right) * Z_{out}$$

Pre-tap continuous time linear equalizer boost is:

$$\left(gm * \frac{Rs}{2} - 1 \right) = k - 2 \quad (18)$$

In some examples, the transistor **206** may be 2/K times the size of the transistor **202**.

FIG. 3 is a schematic diagram of a second example continuous time linear equalizer circuit **300** with pre-tap equalization suitable for use in the equalizer **150**. The continuous time linear equalizer circuit **300** is an example of the continuous time linear equalizer circuit **152**. The continuous time linear equalizer circuit **300** may provide greater pre-tap equalization boost than the continuous time linear equalizer circuit **200**, and may consume less circuit area. The continuous time linear equalizer circuit **300** includes transistors **202**, **204**, **206**, and **208**, capacitor **214**, and current sources **212**, **216**, and **218** similar to the continuous time linear equalizer circuit **200**. The continuous time linear equalizer circuit **300** also includes transistors **302**, **304**, **306**, and **308** that select the connection of the transistor **206** and the transistor **208** to the transistor **202** and the transistor **204**. Pre-tap equalization is selected by turning on the transistor **302** and the transistor **304** (and turning off the transistor **306** and the transistor **308**). Post-tap equalization is selected by turning on the transistor **306** and the transistor **308** (and turning off the transistor **302** and the transistor **304**).

The transistors **302**, **304**, **306**, and **308** may be n-type FETs. The transistor **302** includes a first current terminal (e.g., drain) coupled to the first current terminal of the transistor **204**, and a second current terminal (e.g., source) coupled to the first current terminal of the transistor **206**. A control terminal (e.g., gate) of the transistor **302** is coupled to a pre-tap equalization enable terminal (EN_PRE). The transistor **304** includes a first current terminal (e.g., drain) coupled to the first current terminal of the transistor **202**, and a second current terminal (e.g., source) coupled to the first current terminal of the transistor **208**. A control terminal (e.g., gate) of the transistor **304** is coupled to the control terminal of the transistor **302**.

The transistor **306** includes a first current terminal (e.g., drain) coupled to the first current terminal of the transistor **202**, and a second current terminal (e.g., source) coupled to the first current terminal of the transistor **206**. A control

8

terminal (e.g., gate) of the transistor **306** is coupled to a post-tap equalization enable terminal (EN_PRE). The transistor **308** includes a first current terminal (e.g., drain) coupled to the first current terminal of the transistor **204**, and a second current terminal (e.g., source) coupled to the first current terminal of the transistor **208**. A control terminal (e.g., gate) of the transistor **308** is coupled to the control terminal of the transistor **306**.

The transfer function of the continuous time linear equalizer circuit **300** with pre-tap equalization may be:

$$H_{ctle_pre}(s) = (gm_pre) \left(\frac{1 - SCs * \frac{gm - gm_pre}{gm * gm_pre}}{1 + \frac{S * Cs}{gm}} \right) * Z_{out} \quad (19)$$

Pre-tap equalization boost and transconductance may be defined as:

$$\text{Boost} = \text{Zero/pole} = \left(\frac{gm}{gm_pre} - 1 \right) \quad (20)$$

$$gm_pre = \frac{gm}{1 + gm * Rs} = gm * Rs = k - 1 \quad (21)$$

As per equation (21), the continuous time linear equalizer circuit **300** may provide a pre-tap equalization boost of k-1, rather than k-2 as in the continuous time linear equalizer circuit **200**.

FIG. 4 is a schematic diagram of a third example continuous time linear equalizer circuit **400** with pre-tap equalization suitable for use in the equalizer **150**. The continuous time linear equalizer circuit **400** is an example of the continuous time linear equalizer circuit **152**. The continuous time linear equalizer circuit **400** may provide greater pre-tap equalization boost than the continuous time linear equalizer circuit **200**, and may consume less circuit area. The continuous time linear equalizer circuit **400** includes transistors **202**, **204**, **206**, **208**, and **210**, capacitor **214**, and current sources **212**, **216**, and **218** similar to the continuous time linear equalizer circuit **200**. The continuous time linear equalizer circuit **400** also includes resistors **402** and **404**, and transistor **406**. The transistor **406** connects the resistors **402** and **404** in parallel with the capacitor **214**. A first terminal of the resistor **402** is coupled to the second terminal of the transistor **202**, and a second terminal of the **402** is coupled to a first terminal (e.g., drain) of the transistor **406**. A first terminal of the resistor **404** is coupled to the second terminal of the transistor **204**, and a second terminal of the resistor **404** is coupled to a second terminal (e.g., source) of the transistor **406**. A control terminal (e.g., gate) of the transistor **406** is coupled to pre-tap equalization disable terminal (EN_PREZ). Pre-tap equalization is selected by turning on the transistor **210** and turning off the transistor **406**. Post-tap equalization is selected by turning on the transistor **406** and turning off the transistor **210**.

With pre-tap equalization enabled, the performance of the continuous time linear equalizer circuit **400** is similar to that of the continuous time linear equalizer circuit **300**. With pre-tap equalization disabled (post-tap equalization enabled), the performance of the continuous time linear equalizer circuit **400** is similar to the continuous time linear equalizer circuit **200** in post-top equalization mode. In the continuous time linear equalizer circuit **300**, voltage headroom is consumed by the transistors **302**, **304**, **306**, and **308**.

This causes linearity degradation in the output of the continuous time linear equalizer circuit 300. Accordingly, the continuous time linear equalizer circuit 400 may provide improved linearity relative to the continuous time linear equalizer 300.

FIG. 5 is a schematic diagram of the continuous time linear equalizer circuit 400 showing parasitic resistors 502 and 504 in parallel with the current source 216 and the current source 218. With pre-tap equalization enabled, the resistors 502 and 504 (resistances Rd) across the current sources 216 and 218 draw a current in opposite phase to

$$gm_pre \cdot gmd = \frac{gm}{1 + gmRd}$$

that may degrade equalizer performance.

At low frequencies, the net current is $Gm_dc = (gm_pre - gmd)$. This DC gain is compromised due to the current flowing in the opposite phase. To maintain reasonable DC gain, $Gm_pre_new = gm_pre + gmd$. This restores $Gm_dc = gm_pre$.

At high frequencies, the capacitor 214 bypasses the resistors 502 and 504.

$$Gm_high_freq = gm - gm_pre_new \quad (22)$$

$$Gm_high_freq = gm - gm_pre - gmd \quad (23)$$

$$\text{Boost} = \frac{Gm_high_freq}{Gm_DC} = \frac{gm \cdot gm_pre - gmd}{gm_pre} = K - 1 - \frac{gmd}{gm_pre} \quad (24)$$

$$\text{where } gmd = \frac{gm}{1 + gmRd}, \quad gm_pre = \frac{gm}{1 + gm * Rs / 2},$$

$$\text{Boost} - K - 1 - \frac{1 + gm * \frac{rs}{2}}{1 + gmRd} = K - 1 - RS / 2RD \quad (25)$$

FIG. 6 is a schematic diagram of a fourth example continuous time linear equalizer circuit 600 with pre-tap equalization suitable for use in the equalizer 150. The continuous time linear equalizer circuit 600 is similar to the continuous time linear equalizer circuit 400. The continuous time linear equalizer circuit 600 includes transistors 202, 204, 206, 208, 210, and 406, capacitor 214, resistors 402 and 404, and current sources 212, 216, and 218 similar to the continuous time linear equalizer circuit 400. The continuous time linear equalizer circuit 600 also includes a current source 602, and transistors 604 and 606. The transistors 604 and 606 may be p-type FETs. An input terminal of the current source 602 is coupled to a power supply terminal (VDD). A first current terminal (e.g., source) of the transistor 604 is coupled to an output terminal of the current source 602, and a second current terminal (e.g., drain) of the transistor 604 is coupled to second current terminal of the transistor 204. A control terminal (e.g., gate) of the transistor 604 is coupled to the second current terminal of the transistor 202. A first current terminal (e.g., source) of the transistor 606 is coupled to the output terminal of the current source 602, and a second current terminal (e.g., drain) of the transistor 606 is coupled to second current terminal of the transistor 202. A control terminal (e.g., gate) of the transistor 606 is coupled to the second current terminal of the transistor 204.

The transistors 604 and 606 operate as gds cancellation transistors. The transistors 604 and 606 increase the effective resistance of the resistor 502 and the resistor 504 as:

$$Rd_{eff} = Rd / (1 - Rd * Gm_{can}) \quad (26)$$

where Gm_{can} is the gm of the transistors 604 and 606.

The transistors 604 and 606 reduce the DC output of phase current with respect to gm_pre to produce a pre-tap boost that is very close to $K-1$ in the continuous time linear equalizer circuit 600.

While the continuous linear equalizer circuits 200, 300, 400, and 600 have been described herein as including n-type FETs, some implementations of the continuous linear equalizer circuits 200, 300, 400, or 600 may be implemented using p-type FETs and/or complementary metal oxide semiconductor (CMOS) input structures.

In this description, the term “couple” may cover connections, communications, or signal paths that enable a functional relationship consistent with this description. For example, if device A generates a signal to control device B to perform an action: (a) in a first example, device A is coupled to device B by direct connection; or (b) in a second example, device A is coupled to device B through intervening component C if intervening component C does not alter the functional relationship between device A and device B, such that device B is controlled by device A via the control signal generated by device A.

Also, in this description, the recitation “based on” means “based at least in part on.” Therefore, if X is based on Y, then X may be a function of Y and any number of other factors.

A device that is “configured to” perform a task or function may be configured (e.g., programmed and/or hardwired) at a time of manufacturing by a manufacturer to perform the function and/or may be configurable (or reconfigurable) by a user after manufacturing to perform the function and/or other additional or alternative functions. The configuring may be through firmware and/or software programming of the device, through a construction and/or layout of hardware components and interconnections of the device, or a combination thereof.

As used herein, the terms “terminal,” “node,” “interconnection,” “pin” and “lead” are used interchangeably. Unless specifically stated to the contrary, these terms are generally used to mean an interconnection between or a terminus of a device element, a circuit element, an integrated circuit, a device or other electronics or semiconductor component.

A circuit or device that is described herein as including certain components may instead be adapted to be coupled to those components to form the described circuitry or device. For example, a structure described as including one or more semiconductor elements (such as transistors), one or more passive elements (such as resistors, capacitors, and/or inductors), and/or one or more sources (such as voltage and/or current sources) may instead include only the semiconductor elements within a single physical device (e.g., a semiconductor die and/or integrated circuit (IC) package) and may be adapted to be coupled to at least some of the passive elements and/or the sources to form the described structure either at a time of manufacture or after a time of manufacture, for example, by an end-user and/or a third-party.

While the use of particular transistors is described herein, other transistors (or equivalent devices) may be used instead with little or no change to the remaining circuitry. For example, a field effect transistor (“FET”) (such as an n-channel FET (NFET) or a p-channel FET (PFET)), a bipolar junction transistor (BJT—e.g., NPN transistor or PNP transistor), an insulated gate bipolar transistor (IGBT), and/or a junction field effect transistor (JFET) may be used in place of or in conjunction with the devices described herein. The transistors may be depletion mode devices, drain-extended devices, enhancement mode devices, natural transistors, or other types of device structure transistors. Furthermore, the

11

devices may be implemented in/over a silicon substrate (Si), a silicon carbide substrate (SiC), a gallium nitride substrate (GaN) or a gallium arsenide substrate (GaAs).

References may be made in the claims to a transistor's control input and its current terminals. In the context of a FET, the control input is the gate, and the current terminals are the drain and source. In the context of a BJT, the control input is the base, and the current terminals are the collector and emitter.

References herein to a FET being "ON" or "enabled" means that the conduction channel of the FET is present and drain current may flow through the FET. References herein to a FET being "OFF" or "disabled" means that the conduction channel is not present so drain current does not flow through the FET. An "OFF" FET, however, may have current flowing through the transistor's body-diode.

Circuits described herein are reconfigurable to include additional or different components to provide functionality at least partially similar to functionality available prior to the component replacement. Components shown as resistors, unless otherwise stated, are generally representative of any one or more elements coupled in series and/or parallel to provide an amount of impedance represented by the resistor shown. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in parallel between the same nodes. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in series between the same two nodes as the single resistor or capacitor.

While certain elements of the described examples are included in an integrated circuit and other elements are external to the integrated circuit, in other example embodiments, additional or fewer features may be incorporated into the integrated circuit. In addition, some or all of the features illustrated as being external to the integrated circuit may be included in the integrated circuit and/or some features illustrated as being internal to the integrated circuit may be incorporated outside of the integrated. As used herein, the term "integrated circuit" means one or more circuits that are: (i) incorporated in/over a semiconductor substrate; (ii) incorporated in a single semiconductor package; (iii) incorporated into the same module; and/or (iv) incorporated in/on the same printed circuit board.

Uses of the phrase "ground" in the foregoing description include a chassis ground, an Earth ground, a floating ground, a virtual ground, a digital ground, a common ground, and/or any other form of ground connection applicable to, or suitable for, the teachings of this description. In this description, unless otherwise stated, "about," "approximately" or "substantially" preceding a parameter means being within ± 10 percent of that parameter or, if the parameter is zero, a reasonable range of values around zero.

Modifications are possible in the described embodiments, and other embodiments are possible, within the scope of the claims.

What is claimed is:

1. A circuit comprising:

a first transistor having a first terminal, a second terminal, and a control terminal;

a second transistor having a first terminal, second terminal, and a control terminal;

a capacitor having a first conductor coupled to the second terminal of the first transistor, and a second conductor coupled to the second terminal of the second transistor;

12

a third transistor having:

a first terminal coupled to the first terminal of the second transistor;

a second terminal; and

a control terminal coupled to the control terminal of the first transistor; and

a fourth transistor having:

a first terminal coupled to the first terminal of the first transistor;

a second terminal coupled to the second terminal of the third transistor; and

a control terminal coupled to the control terminal of the second transistor.

2. The circuit of claim 1, further comprising:

a fifth transistor having:

a first terminal coupled to the second terminal of the fourth transistor;

a second terminal; and

a control terminal coupled to a pre-tap equalization enable terminal; and

a current source having:

a first terminal coupled to the second terminal of the fifth transistor; and

a second terminal coupled to a reference voltage terminal.

3. The circuit of claim 1, further comprising:

a resistor having:

a first terminal coupled to the second terminal of the first transistor; and

a second terminal coupled to the second terminal of the second transistor;

a first current source having:

a first terminal coupled to the second terminal of the first transistor; and

a second terminal coupled to a reference voltage terminal; and

a second current source having:

a first terminal coupled to the second terminal of the second transistor; and

a second terminal coupled to a reference voltage terminal.

4. The circuit of claim 1, further comprising:

a fifth transistor having a first terminal, a second terminal, and a control terminal the control terminal coupled to a pre-tap equalization disable terminal;

a first resistor having:

a first terminal coupled to the second terminal of the first transistor; and

a second terminal coupled to the first terminal of the fifth transistor;

a second resistor having:

a first terminal; and coupled to the second terminal of the fifth transistor; and

a second terminal coupled to the second terminal of the second transistor;

a first current source having:

a first terminal coupled to the second terminal of the first transistor; and

a second terminal coupled to a reference voltage terminal; and

a second current source having:

a first terminal coupled to the second terminal of the second transistor; and

a second terminal coupled to the reference voltage terminal.

13

5. The circuit of claim 3, further comprising:
 a fifth transistor having:
 a first terminal;
 a second terminal coupled to first terminal of the
 second current source; and
 a control terminal coupled to the second terminal of the
 first transistor;
 a sixth transistor having:
 a first terminal coupled to the first terminal of the fifth
 transistor;
 a second terminal coupled to the first terminal of the
 first current source; and
 a control terminal coupled to the second terminal of the
 second transistor; and
 a third current source having:
 a first terminal coupled to a power supply terminal; and
 a second terminal coupled to the first terminal of the
 fifth transistor.
6. The circuit of claim 1, further comprising:
 a fifth transistor having:
 a first terminal coupled to the first terminal of the
 second transistor;
 a second terminal coupled to the first terminal of the
 third transistor; and
 a control terminal;
 a sixth transistor having:
 a first terminal coupled to the first terminal of the first
 transistor;
 a second terminal coupled to the first terminal of the
 fourth transistor; and
 a control terminal coupled to the control terminal of the
 fifth transistor;
 a seventh transistor having:
 a first terminal coupled to the first terminal of the sixth
 transistor;
 a second terminal coupled to the second terminal of the
 fifth transistor; and
 a control terminal;
 an eighth transistor having:
 a first terminal coupled to the first terminal of the fifth
 transistor;
 a second terminal coupled to the second terminal of the
 sixth transistor; and
 a control terminal coupled to the control terminal of the
 seventh transistor.
7. A circuit comprising:
 a first transistor having a first terminal, a second terminal,
 and a control terminal;
 a second transistor coupled to the first transistor as a first
 differential pair, the second transistor having a first
 terminal, a second terminal, and a control terminal;
 a third transistor having:
 a first terminal coupled to the first terminal of the
 second transistor;
 a second terminal; and
 a control terminal coupled to the control terminal of the
 first transistor;
 a fourth transistor coupled to the third transistor as a
 second differential pair, the fourth transistor having:
 a first terminal coupled to the first terminal of the first
 transistor;
 a second terminal; and
 a control terminal coupled to the control terminal of the
 second transistor; and
 a fifth transistor coupled to the second differential pair, the
 fifth transistor configured to switchably enable pre-tap
 equalization.

14

8. The circuit of claim 7, further comprising:
 a current source having:
 a first terminal; and
 a second terminal coupled to a reference voltage ter-
 minal;
 wherein:
 the fifth transistor has:
 a first terminal coupled to the second terminal of the
 third transistor and the second terminal of the
 fourth transistor;
 a second terminal coupled to the first terminal of the
 current source; and
 a control terminal coupled to a pre-tap equalization
 enable terminal.
9. The circuit of claim 7, further comprising:
 a sixth transistor having:
 a first terminal coupled to the first terminal of the first
 transistor;
 a second terminal coupled to the first terminal of the
 fourth transistor; and
 a control terminal;
 a seventh transistor having:
 a first terminal coupled to the first terminal of the sixth
 transistor;
 a second terminal coupled to the first terminal of the
 third transistor; and
 a control terminal;
 an eighth transistor having:
 a first terminal coupled to the first terminal of the
 second transistor;
 a second terminal coupled to the second terminal of the
 sixth transistor; and
 a control terminal coupled to the control terminal of the
 seventh transistor;
 wherein the fifth transistor has:
 a first terminal coupled to the first terminal of the
 second transistor;
 a second terminal coupled to the first terminal of the
 third transistor; and
 a control terminal coupled to the control terminal of the
 sixth transistor.
10. The circuit of claim 7, further comprising a capacitor
 having:
 a first conductor coupled to the second terminal of the first
 transistor; and
 a second conductor coupled to the second terminal of the
 second transistor.
11. The circuit of claim 10, further comprising:
 a first current source having:
 a first terminal coupled to the first conductor;
 a second terminal coupled to a reference voltage ter-
 minal; and
 a second current source having:
 a first terminal coupled to the second conductor; and
 a second terminal coupled to the reference voltage
 terminal.
12. The circuit of claim 7, further comprising a resistor
 having:
 a first terminal coupled to the second terminal of the first
 transistor; and
 a second terminal coupled to the second terminal of the
 second transistor.
13. The circuit of claim 7, further comprising:
 a first resistor having:
 a first terminal coupled to the second terminal of the
 first transistor; and
 a second terminal;

15

a second resistor having:
 a first terminal coupled to the second terminal of the second transistor; and
 a second terminal;
 a sixth transistor having: 5
 a first terminal coupled to the second terminal of the first resistor;
 a second terminal coupled to the second terminal of the second resistor; and
 a control terminal coupled to a pre-tap equalization disable terminal. 10

14. The circuit of claim **13**, further comprising:
 a current source having:
 a first terminal coupled to a power supply terminal; and 15
 a second terminal;
 a seventh transistor having:
 a first terminal coupled to the second terminal of the current source;
 a second terminal coupled to the second terminal of the second transistor; and 20
 a control terminal coupled to the second terminal of the first transistor; and
 an eighth transistor having:
 a first terminal coupled to the second terminal of the current source; 25
 a second terminal coupled to the second terminal of the first transistor; and
 a control terminal coupled to the second terminal of the second transistor. 30

15. A deserializer comprising:
 a clock data recovery circuit;
 a linear equalizer circuit coupled to the clock data recovery circuit, the linear equalizer circuit including:
 a first transistor having a first terminal, a second 35
 terminal, and a control terminal;
 a second transistor having a first terminal, second terminal, and a control terminal;
 a capacitor having:
 a first conductor coupled to the second terminal of 40
 the first transistor; and
 a second conductor coupled to the second terminal of the second transistor;
 a first current source having:
 a first terminal coupled to the first conductor; and 45
 a second terminal coupled to a reference voltage terminal;
 a second current source having:
 a first terminal coupled to the second conductor; and
 a second terminal coupled to the reference voltage 50
 terminal;
 a third transistor having:
 a first terminal coupled to the first terminal of the second transistor;
 a second terminal; and 55
 a control terminal coupled to the control terminal of the first transistor; and
 a fourth transistor having:
 a first terminal coupled to the first terminal of the first transistor; 60
 a second terminal coupled to the second terminal of the third transistor; and
 a control terminal coupled to the control terminal of the second transistor; and
 a fifth transistor coupled to the third transistor, the fifth 65
 transistor configured to switchably enable pre-tap equalization.

16

16. The deserializer of claim **15**, wherein:
 the linear equalizer circuit includes:
 a current source having:
 a first terminal; and
 a second terminal coupled to a reference voltage terminal; and
 the fifth transistor has:
 a first terminal coupled to the second terminal of the fourth transistor;
 a second terminal coupled to the first terminal of the current source; and
 a control terminal coupled to a pre-tap equalization enable terminal.

17. The deserializer of claim **15**, wherein:
 the fifth transistor has:
 a first terminal coupled to the first terminal of the second transistor;
 a second terminal coupled to the first terminal of the third transistor; and
 a control terminal; and
 the linear equalizer circuit includes:
 a sixth transistor having:
 a first terminal coupled to the first terminal of the first transistor;
 a second terminal coupled to the first terminal of the fourth transistor; and
 a control terminal coupled to the control terminal of the fifth transistor;
 a seventh transistor having:
 a first terminal coupled to the first terminal of the sixth transistor;
 a second terminal coupled to the second terminal of the fifth transistor; and
 a control terminal;
 an eighth transistor having:
 a first terminal coupled to the first terminal of the fifth transistor;
 a second terminal coupled to the second terminal of the sixth transistor; and
 a control terminal coupled to the control terminal of the seventh transistor.

18. The deserializer of claim **15**, wherein the linear equalizer circuit includes a resistor having:
 a first terminal coupled to the first conductor of the capacitor; and
 a second terminal coupled to the second conductor of the capacitor.

19. The deserializer of claim **15**, wherein the linear equalizer circuit includes:
 a first resistor having:
 a first terminal coupled to the second terminal of the first transistor; and
 a second terminal;
 a second resistor having:
 a first terminal coupled to the second terminal of the second transistor; and
 a second terminal;
 a sixth transistor having:
 a first terminal coupled to the second terminal of the first resistor;
 a second terminal coupled to the second terminal of the second resistor; and
 a control terminal coupled to a pre-tap equalization disable terminal.

20. The deserializer of claim **15**, wherein the linear equalizer circuit includes:

17

- a third current source having:
a first terminal coupled to a power supply terminal; and
a second terminal;
- a sixth transistor having:
a first terminal coupled to the second terminal of the 5
third current source;
a second terminal coupled to the first terminal of the
second current source; and
a control terminal coupled to the first terminal of the
first current source; and 10
- a seventh transistor having:
a first terminal coupled to the second terminal of the
third current source;
a second terminal coupled to the first terminal of the
first current source; and 15
- a control terminal coupled to the first terminal of the
second current source.

* * * * *

18